

Chapter 14.0: Presentation of the Spring Course

Chapters 1-13 of the study material are related to the Programming 1 course in last autumn. The spring course Programming Studio A continues from it and covers chapters 14-19.

About this page:

Questions Answered: What is the course Programming Studio 2 about? How does the course work?

Topics: Course objectives, parts, studying methods, assessment, information and counselling channels.

What Will I Do? Read, respond to background survey

Point value: 5

Indicative difficulty level: Easy.

Rough Estimate of Workload:? Half an hour.

Related Projects: None.

MOOC continues, In Aalto the course and semester change

Welcome to the Programming Studio A course! The course continues from where Programming 1 ended. (chapter 14) The main goal of this course is to learn the skills needed to design and implement a program from scratch, and to practice this by implementing a reasonably large personal programming project.

The Finnish version of the course, CS-C2120 Programming Studio 2, is for students in the Finnish-speaking Computer Science bachelor program. The two courses share most of the course resources and exercises. The key difference is in the set of offered project topics and a demo session organized on Fridays.

Déjà vu?

If you feel you've read some of this before, then you're partially right. Spring and autumn courses have a lot in common. Important new things for Aalto students who continue straight from the autumn course are enrolling in Sisu, visiting MyCourses, getting to know the course schedule and how the grade of the course is formed. Sisu and MyCourses are only relevant during the course as some mass postings are carried out via MyCourses, and the feedback survey will be sent to those registered to Sisu. Sisu registration is also needed for registering the course grades. MOOC students do not have to worry about these, but of course, course schedule and grading information are important for them, too.

This material is offered in A-plus as an ebook combining learning resources and exercises. In case we refer to chapters 1-13 these are part of the course Programming 1, you most likely partook in autumn.

MyCourses site

In addition to the A-plus system, Aalto's general MyCourses website is also in use. However, the material offered by our online course is so varied that it is difficult to present it in MyCourses. Therefore it is here in A-Plus. On the MyCourses page you will find mainly:

- a brief course description
- the most important announcements of the course (these are also published in A-plus course news)
- the times and places of the exercise groups
- lecture slides, lecture recordings and other possible resources covered in lectures.

It is very important that you follow the news either in A-plus or MyCourses. These contain information about the current issues during the course such as course deadline reminders or unforeseen changes, such as exemptions in staffing or arrangements. In case of technical problems, which might happen during the course in A-plus system or other IT problems preventing working with the exercises, we announce delays of deadlines in both platforms if possible.

Registration

If you are an Aalto student, you should have registered to the course in Sisu. We do however collect more background information about the course participants, for adjusting the course to better match its expected audience. Please respond to the survey at the end of this page. Thank you!

Both Aalto and MOOC students can begin the course immediately by logging in to the A-plus system with your Aalto / MOOC credentials. There should be a registration button on the main page of the course.

Course Objectives

Overall, the course objectives are, as follows.
This course has already ended.

After completing the course ...

- you can design and implement a small object-oriented program with several classes and traits
- you can compare, contrast and discuss different designs
- you have extended your skills about various program structures in Scala and in general,
- and you have improved your skills in implementing and testing programs, as well as using different programming tools to support the programming process

The objective of the course is therefore to learn how to design, implement and modify small application programs such as games, book-keeping software or small simulations. In addition, we hope you find that learning these skills can be both fun and useful! The goals also include familiarising yourself with various programming support tools/methods, such as version control, build tools and testing.

Some topics have been briefly covered in the Programming 1 course; however, this course extends and deepens what you learned in the autumn course. It also opens up opportunities to participate in new courses that are more focused on a particular topic or participating in a larger group project. You will learn about the continuation courses at the end of this course.

In addition to the above, you will familiarise with some new programming structures and techniques that we did not discuss in the autumn.

Note. If you have taken the course Programming 1 earlier than Autumn 2021 and have not programmed with Scala thereafter, we recommend you to revisit Scala programming before starting to work with exercises of this course.

Prerequisites

Course participants are assumed to have completed the course Programming 1, and most likely the major part of B level assignments. If you completed mainly A level assignments, you are likely to face problems in this course.

These course resources are targeted to Programming Studio A. The same resources in Finnish are targeted to the course CS-C2120, Programming studio 2.

Some resources are available to both courses in English.

Course activities

The course consists of four parts, the first two being linked together:

1. **Exercise tasks** are related to the content to be studied and support your ability to experiment with the skills required in the project. The assignments are given a sub-grade that, for the MOOC and Aalto students, becomes a part of the final grade in different ways. Passing the course requires an approved grade for the

exercises, ie. at least 1. The weight of the assignments is 30% of the final grade for Aalto students.

2. At the end of each section of the study material we collect **section feedback**, that is, questions about the section contents, opinions on the material, reflections on learning, etc. Feedback on time usage / complexity is a compulsory part of the course! Results of the assignments will not be confirmed until the small feedback form at the end of the section has been completed. In most of the sections, it is however enough to give us feedback on the time you spent on working with the section (reading and solving the assignments).
3. At the end of the course, you must complete a **course feedback survey**, which evaluates the course as a whole and is published as part of the course material. Responding to this questionnaire is mandatory.
4. Aalto students (but not MOOC students) also have a **personal programming project**. The project starts alongside with the exercises in the beginning of February by selecting a project topic. The project plan is developed alongside the exercises, and it is presented to the course assistants approximately on weeks 8-9. After the assistant has approved the plan, you begin project implementation which takes place during period IV and the beginning of period V (weeks 10-17). The same assistant who inspects the plan will also follow the progress of the project throughout the project. The project, with its plans, will be stored in a version control system (also learned as part of the course), through which the assistant will also follow your reports on how the project progresses. Assistants will organize one or more personal meetings with you depending on the project progress. The final project and its document are submitted by April 27th, and the working program will be demonstrated to the assistant late April or early May. The weight of the project is 70% of the Aalto students' final grade.

Even though project planning is already underway during the exercises, the project does not affect the activities of the weekly exercise rounds and is not part of their scoring.

Below you will find more about assignments and section-specific feedback. The course feedback survey will be announced separately in MyCourses news section and Piazza later in spring.

Final grade calculation

The MOOC students only carry out the exercise assignments. Based on their results, students can be awarded a 2-credit course mark. Those who did not /cannot apply for FiTech-credits, can download/request a diploma from the course web site similarly as on the autumn course.

Aalto students complete the course by solving the weekly exercises, and the personal project, including project plan, required reports/documents and demos. The course final grade is based on the weighted average of the grade of the exercises and the grade of the project. The weight of the exercises is 30%, the project is 70%. The project is graded based on project plan, interim reporting, quality of the submitted program and its code, as well as final report. The project evaluation criteria are described in more detail on the project pages.

Exercises

The exercises of the course are embedded in this learning material. You will carry out them while learning about the reading material. There is a **corresponding exercise** round in Chapters 14-19, and each round includes a variety of programming exercises and multiple-choice questions which test understanding some topic.

Exercise rounds

Assignments are divided into rounds corresponding to Chapters 14-19. Each round has a *deadline* (DL), due which the solutions must be submitted. The first round DL is Fri 21.1.2022 18:00. The deadlines for the next rounds are usually one week from the previous one. The exercises rounds are published roughly two weeks before the deadline, which imply that the rounds overlap. Aalto students make the project plan during the final rounds, thus there are some less tasks in the last rounds.

Deadline!

The first DL is on 21.1. 18:00 Note that the first round of the spring will take some time. It is worth reserving enough time for the assignment!

Score

The grade limits below will be adjusted in January after all course exercises have been installed in A+.

At the beginning of each of section there is an indication of its point value: the letter S (studio) and the number, eg. "S50". The letter S separates chapter 14- points from the rounds of the autumn and there is no substitute between these points in either direction

Following the letter "S" is the maximum score that can be obtained from the total section.

The grade of the course exercise part (0-5) depends on the number of points received from the exercises, according to the following table. For MOOC students the grade is also the final grade of the MOOC spring part.

Exercise grade	Level S	Description
1	620	Minimum requirements achieved
2	780	
3	900	
4	1050	
5	1200	No more is needed for grade 5.
maximum	1395	You do not need to get <i>all</i> points.

(In addition, Aalto students need to respond to the course feedback questionnaire in May.)

Below there is a table of points available in each round.

Round	Level S
14	255
15	260
16	340
16	110
18	360
19	70
All	1395

Note. The total maximum points visible in A+ will change during the course, because all new exercises have not yet been configured in A+.

Submitting exercises and automatic assessment

The study material includes both text and assignments.

In some of the tasks, you can directly answer the question in the browser when reading the material. For some of the tasks, you write the answer outside the browser to a separate file or files. The latter tasks are usually programming exercises where you create or edit an existing program. These tasks are also submitted in the same way as in the autumn course, because this course will use IntelliJ IDE.

For each programming assignment, you get a number of points, primarily based on whether the submitted program works as specified in the assignment. The assessment system reviews almost all tasks automatically and provides immediate feedback on the submitted solutions. This inspection is rather picky, so you have to be careful. On the other hand, the evaluation is available for 24h / day, fast and fair.

In some assignments, teaching assistants evaluate the solutions. This is mentioned in the assignments separately.

Most of the tasks have a certain limited number of allowed resubmissions. For the programming tasks which are returned in separate files, this number is usually ten. For small multiply choice or equivalent tasks the number of submissions is three to five. The best result achieved before the deadline will remain in effect, so your points will never decrease.

Model solutions for the assignments are published after each round has been closed. Note that there is no model solution for the project plan.

Get ready to work in time!

The vast majority of exercise points are gained from tasks where you program yourself. You may need to use considerably time on these tasks. Please note that it is not advisable to leave working with the programs till the day before the deadline, or your working time plan could be ruined. Take a look at the assignments in advance, create a time plan for yourself, and start working as early as possible.

Confirmation of exercise points and section feedback

There is a feedback form at the end of this page and correspondingly in all other course sections. *Answering these read-only forms is a compulsory part of the course: the exercise points from the section are not recorded before you answer the section-based feedback form!*

Working with sections happens like this:

- You read the section and do the related tasks. A-plus evaluates your response and announces how much points you will get.
- You complete a read-only feedback form. *Writing the feedback has the same deadline as the round anyway!*
- The course staff will read your feedback within a few days and will register it as approved. The task points you get from the section will then be formally recorded. In most sections, verbal feedback is voluntary and we are primarily interested in how much time you needed to work with the section.

If verbal feedback is required, then the form will be accepted as long as there is something reasonable in the free-form feedback box. "Reasonable" may be, for example, a question about the content of the course, a comment on the quality of the score, a brief reflection on one's own learning, etc. Nothing extensive writing is required. More detailed instructions can be found in the form.

Feedback is taken into account!

We're not asking you questions and comments just to get rid of them! We will read all of them all and use the information during the course and when planning the future course versions.

Pair programming

Exercises are meant to be solved *in pairs*.

It is, of course, permitted to solve them solo if you want. However, when co-operating it is imperative to communicate with the pair which helps to structure the ideas better. Conversation is one important form of learning. One finds the mistakes that others missed. Social pressure also prevents too much laziness. However, in pair working it is essential that both carry out programming activities, because programming cannot be learned without doing it oneself.

On the other hand, the *project* and all its phases will be done alone. It is allowed to discuss the project and programming problems with others as before, but each student is supposed to plan and implement his own project independently.

Stating the obvious

Copying solutions from other students or elsewhere is not permitted. Likewise, it is forbidden to obtain points by trying to deceive the automated control system in some way.

Copying and other abuses have consequences 
(<https://into.aalto.fi/display/ensaannot/Aalto+University+Code+of+Academic+Integrity+and+Handling+Violations+Thereof>),
Submitted programs will also be studied in this regard.

To support the study: exercise groups, forums, etc.

Each student's own effort is central to learn the material and solve the involved tasks. However, we support learning in many different ways. There are exercise groups, a few lectures and electronic discussion forums. Based on the feedback some additional resources can be generated in terms of weekly summaries.

Lectures

The course has one lecture per week on period III. Lecture slides and other lecture material will be published in MyCourses. Lectures are given in English on Wednesdays 12-14 and in Finnish for Studio 2 course on Wednesdays 14-16.

In addition to this, there is an additional session on Fridays 12-14 to deepen the topics covered in the same week's lecture, for example with the help of practical examples. Friday's event will be held in English and is common to the courses Studio 2 and Studio A.

The opening lecture in English will be on Wednesday 12.1.2022. It introduces the course practices, timetable, etc. In addition, some topics of the first round (Chapter 14, program design) are covered.

All lectures are given at zoom address

<https://aalto.zoom.us/j/65973501783>  (<https://aalto.zoom.us/j/65973501783>)

The lectures are recorded and stored in the Aalto Panopto system at latest one day after the lecture. we welcome audience questions and you can keep your camera on or set a profile picture to make the lecture feel more like a normal event.

Exercise Groups

The course assistants are on duty **in exercise groups** during Period 3. Because the university will operate in distance learning mode in spring 2022 due to a pandemic, the exercises are organized as zoom video meetings. We might transition to hybrid mode as soon as its feasible.

All exercise groups take place in zoom room

<https://aalto.zoom.us/j/61838001876>  (<https://aalto.zoom.us/j/61838001876>)

Teaching assistants attend the room and when appropriate, discussions are moved into separate breakout rooms.

Exercise groups start on Mon 17.1. **In period III**, two four hour sessions and one two hour session will be organized weekly. You are free to join in anytime during the session to ask for advice.

- Mondays 14:15-18:00,
- Tuesdays 12:15-16:00 and
- Fridays 14:15-16:00.

The times are also available in MyCourses.

In **Period IV and in beginning of Period V**, the exercise groups are on

- Mondays 14.15-16, and
- Thursdays, 12.15-14

At this time the sessions are for project work support. It is likely that classrooms are in normal use and many students find the classroom to be a good place to concentrate and to ask for assistant help when needed.

Exercise groups are voluntary and you do not need to enrol in them; you can visit them freely. Nor do the groups have a specific agenda, so that all students would work at the same pace. You can come to the group to read the material and perform tasks at your own pace and ask the assistants when you do not know how to proceed.

Discussion forums: Zulip

Aalto is changing the services that are used to implement discussion forums. The course will test the Zulip service. You can log in to it at

<https://studios.zulip.aalto.fi/>  (<https://studios.zulip.aalto.fi/>)

Each chapter in the course has its own discussion stream where you can seek advice from staff or other students. Answers are received usually in no later than 24 hours. You can also discuss other course related questions here.

About email

Please, do not ask your assistants for advice with programming assignments by email. The majority of the assistants are students who work only a few hours per week and they have their own studies, too. Ask for advice in the exercise groups or in the electronic forums. The course has a lot of channels for programming questions, but e-mail does not belong to them.

Staff

University lecturer Otto Seppälä

Email: otto.seppala@aalto.fi (mailto:otto.seppala%40aalto.fi)

Juha Sorva, the Lecturer of the Fall course who has completed the course material chapters 1-13 is supporting the course by providing some feedback and insights, but he is not participating in course teaching.



(https://gitmanager.cs.aalto.fi/static/studio2_k2022dev2-horrible-gitmanager-interface/_images/otto_seppala.jpg)

Assistants

In addition to the aforementioned, the staff of the course includes teaching assistants, who help students in the exercise groups, and provide also advice in the discussion forum. They also evaluate some of the exercises and perform various other tasks. They also guide and evaluate projects.

In spring 2022 the teachings assistants are:

The open MOOC course

The course is also given as a purely online version (MOOC), which is open to anyone other than Aalto's own students (e.g. high school students and teachers).

Welcome again!

We, the course staff, hope to make the course successful. We also hope you will invest in working on the course, even though it is laborious and challenging!

Please, respond to the following questions.

⚠ This course has been archived (Thursday, 1 June 2023, 12:00).

You cannot submit this assignment

Background survey

This course has been archived (Thursday, 1 June 2023, 12:00).

* Question 1 0 / 0

What is the highest degree that you have completed?

- Show anyway
- ☐ I have not completed any of these degrees
 - ☐ Primary or lower secondary school
 - ☒ **Matriculation examination (upper secondary school, high school) or a vocational examination**
 - ☐ Associate degree (in Finland, this degree was awarded only before the reformation of the educational system)
 - ☐ (Bachelor's) degree from a university of applied sciences or polytechnic
 - ☐ Bachelor's degree from a university
 - ☐ Master's degree from a university
 - ☐ Doctor's or Licentiate's degree from a university

✘ Incorrect

*** Question 2 0 / 0**

Which of the following alternatives describes best your status as a student?

- ☒ **I study in one of the Bachelor's programmes of Aalto University (I have not a Bachelor's degree yet).**
- ☐ I study in one of the Master's programmes of Aalto University (I have completed a Bachelor's degree).
- ☐ I am a doctoral student at Aalto University.
- ☐ I enrolled in this course as a student of the FiTech network.
- ☐ I am a student of Open University.
- ☐ I am a degree student in another university and enrolled in this course through JOO agreement.
- ☐ I am an exchange student in Aalto University.
- ☐ I am a high school student and my high school offers me an opportunity to take this course.
- ☐ None of the above alternatives describes me.

✖ Incorrect

Question 3 0 / 0

If you are the degree student in one of the Bachelor's programmes of Aalto University, through which study option did you enter the university?

Aalto Bachelor's Programme in Science and Technology - Data Science ▼

✖ Incorrect

Question 4 0 / 0

If you are a degree student in one of the Bachelor's programmes of Aalto University (i.e., you have not a Bachelor's degree yet), which is your class year (count only the years you have been enrolled as an active student)?

- ☒ **This is my first year.**
- ☐ This is my second year.
- ☐ This is my third year.
- ☐ This is my fourth year.
- ☐ This is at least my fifth year.

✖ Incorrect

*** Question 5 0 / 0**

Read through the document Aalto University Code of Academic Integrity and Handling Violations Thereof (<https://into.aalto.fi/display/ensaannot/Aalto+University+Code+of+Academic+Integrity+and+Handling+Violations+Thereof>).

- ☒ **I understand the Code of Academic Integrity and I am committed to following them in my studies.**

✖ Incorrect

Submit

Summary

- In spring 2022, the course is fully organised online.
- The compulsory parts of the course are: exercise assignments, section feedback / time estimates, all phases of the project, and the course feedback at the end.
- Studying in the course material is worthwhile. You will need the provided information both in the exercises embedded in the material and in the course

project. The final grade of the course is calculated as a weighted average of the exercises and project grades.

- 30% of the course grade will be based on the exercises, as long as the feedback sections are submitted. The first feedback section is below.
- 70% of the grade is based on the project, which includes project plan and its demo, use of version control, interim reporting, final working project with document and project demo.
- Exercise groups, discussion forums, lectures, and weekly summaries support your work.
- The first deadline is soon!
- If you want, find yourself a pair for the exercise part of the course.

Feedback

Please note that this section must be completed individually. Even if you worked on this chapter with a pair, each of you should submit the form separately

 This course has been archived (Thursday, 1 June 2023, 12:00).

You cannot submit this assignment

Time spent: (*) Pakollinen Please estimate the total number of minutes you spent on this chapter (reading, assignments, etc.). You don't have to be exact, but if you can produce an estimate to within 15 minutes or half an hour, that would be great.

10

Show anyway

"I feel that I have understood the most important things in this chapter." (*) Pakollinen

- ☒ fully agree
- ☐ somewhat agree
- ☐ somewhat disagree
- ☐ fully disagree
- ☐ I'm unable to answer or don't want to comment.

Submit an update

